Jodh Singh Nahal

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Education

University of British Columbia *Vancouver, BC*

Bachelor of Applied Science in Computer Engineering *Expected May 2028*

- Presidential Scholars Award ($60,000), Dean's Honour List, available 4 or 8 months, CGPA: 84.7%.

Technical Skills

Languages: C/C++, Python, ARM Assembly, SQL, Shell
Performance: Multi-threading, Non-Blocking Architecture, Code Profiling, Low-Latency Systems
Tools: Linux, GDB, Git, Docker, Jenkins, CMake

Experience

Hardware Verification Engineering Intern *Vancouver, BC*

AMD *Sept 2026 – Present*

- Building automation scripts and regression workflows to improve validation efficiency and coverage closure.
- Debugging RTL behavior through simulation and waveform analysis.
- Contributing to verification of the IOHUB subsystem for next-generation AMD processor architectures.
- Verifying IOMMU L1 cache behavior using UVM in a constrained-random testbench.
- Expanded the constrained-random testbench to cover corner cases and improve functional coverage.

Software Test / Embedded Development Co-op *Burnaby, BC*

Xantrex *May 2024 – Aug 2025*

- Developed an object-oriented, multi-threaded, non-blocking C++ firmware architecture on Linux-based Raspberry Pi gateways to parse RV-C CAN frames from STM32 devices and publish structured messages over MQTT.
- Achieved 7× device capacity, 90% lower CPU usage, and ∼2 s lower latency after the C++ migration.
- Reduced build times by 5× by implementing CI/CD cross-compilation pipelines with Docker, Jenkins, and CMake.
- Built Python Hardware-in-the-Loop (HIL) CAN and UART simulators to run 100+ verification tests on the Raspberry Pi gateway, validating edge cases and stress-testing real-time performance.
- Replaced a ∼60k-line Python implementation with the C++ architecture.
- Wrote Python scripts to autogenerate scalable C++ object-oriented code from SQLite and Microsoft Access/Excel data, eliminating manual boilerplate.
- Profiled the C++ firmware to eliminate performance bottlenecks.

Firmware / Embedded Systems Developer *Vancouver, BC*

UBC Rover *Sept 2023 – July 2026*

- Automated firmware validation with GitHub Actions CI and Catch2 unit tests to improve regression coverage.
- Led a team of 3 engineers building a distributed embedded control system across ESP32 (FreeRTOS), STM32, Arduino, and TI LaunchPad MCUs over SPI/I2C.
- Aggregated inputs from switches, buttons, potentiometers, encoders, and joysticks and streamed data over UART.

Undergraduate Research Assistant *Vancouver, BC*

UBC Electrical and Computer Engineering — Supervisor: Dr. Lutz Lampe *May 2026 – Sept 2026*

- Optimizing power extraction of piezoelectric energy harvesting interface circuits (SEH, SSHI, SECE) through LTspice and Simulink/Simscape simulation.
- Researching battery-less platforms for the Internet of Underwater Things via simulation of piezoelectric interface circuits.

Projects

Memory Allocator Implementation | *C, Pointers, Systems Programming*

- Developed a custom malloc, free, and realloc using an implicit free list with 8-byte alignment and block headers.
- Managed block metadata, fragmentation, and coalescing with pointer arithmetic for efficient heap utilization.

OS161 Kernel Development | *C, GDB, MIPS, Operating Systems*

- Engineered thread synchronization primitives, including blocking locks and condition variables, to resolve race conditions in kernel space.